higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE

MATHEMATICS N5

(16030175)

30 July 2026 (X-paper)
09:00–12:00

Drawing instruments and scientific calculators may be used.

This question paper consists of 5 pages and a formula sheet of 4 pages.

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DEPARTMENT OF HIGHER EDUCATION AND TRAINING
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE
MATHEMATICS N5
TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer all the questions.
2. Read all the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Show all intermediate steps and simplify where possible.
5. All final answers must be rounded off to THREE decimal places.
6. Questions may be answered in any order, but subsections of questions must be kept together.
7. Sketches must be large, neat and fully labelled.
8. Scientific calculator may be used.
9. Use only a black or blue pen.
10. Write neatly and legibly.

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QUESTION 1

1.1 Determine the following limit:

1.1.1 $y = \lim_{x \to 2} \left(\frac{1}{x-2} - \frac{1}{x^2 - 3x + 2} \right)$ (2)

1.1.2 $y = \lim_{x \to \infty} \frac{\ln(x+2)}{x \log_2 x}$ (3)

1.2 Determine the value(s) of x for which $f(x)$ is discontinuous if:

$$f(x) = \frac{x^2 + 5x - 24}{x^2 - x - 6}$$ (2)

[7]

QUESTION 2

2.1 Given:

$$f(x) = \frac{1}{4x^4}$$

Determine the derivative of $f(x)$ from first principles. (5)

2.2 Determine $\frac{dy}{dx}$ in each of the following cases:

(Simplification NOT required)

2.2.1 $y = \text{cosec}^4(\tan 2x)$ (4)

2.2.2 $y = \frac{\sqrt[3]{x} - 2xe^x}{(3x+1)}$ (4)

2.3 Calculate $\frac{dy}{dx}$ if $y = (\sec 3x)^{4x^2}$ with aid of logarithmic differentiation. (4)

2.4 Given the implicit function: $\sec(x - y) = y \,\text{cosec}\, x$

Determine $\frac{dy}{dx}$ (5)

[22]

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QUESTION 3

3.1 Given: $f(x) = 3x^3 - 4x + 3$

3.1.1 Determine the coordinates of the turning points of $f(x)$. (3)

3.1.2 Draw up a table of x and $f(x)$, where x is ranging from:

$x = -2$ to $x = 2$; $x \in Z$. (2)

3.1.3 Draw a neat graph of $f(x)$ between these values showing the turning points (2)

3.1.4 Given that an estimate of the root of the equation $3x^3 - 4x + 3 = 0$ is close to $x = -1{,}3$, use Taylor's /Newton's method TWICE to determine a better approximation of this root.

(Correct root to THREE decimal figures.) (4)

3.2 You have 100 metres of fence to make a rectangular play area alongside the wall of your house. The wall of the house bounds one side.

What is the largest size possible (in square metres) for the play area? (6)

3.3 The height (in meters) after t seconds, of a projectile shot vertically upwards from a point 2 m above the ground level with an initial 24.5 m/s is $h = 2 + 24{,}5t - 4{,}9t^2$.

How long will projectile take to reach maximum height? (3)

[20]

QUESTION 4

4.1 Determine $\int y\,dx$ in each of the following cases:

4.1.1 $y = \dfrac{\tan x \sec^2 x}{1 - \tan^2 x}$ (3)

4.1.2 $y = \cos^2 6x$ (3)

4.1.3 $y = \sin(3x)\sin\left(\dfrac{4}{3}x\right)$ (3)

4.1.4 $y = \dfrac{x^3 - 3x + 2}{x + 1}$ (5)

4.1.5 $y = 2\sin x \ln(\sec x)$ (4)

4.2 Determine $\int y\,dx$ by resolving the integral into partial fractions:

$y = \dfrac{x - 7}{x^2 + x - 2}$ (5)

[23]

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QUESTION 5

5.1 Determine:

$$\int_0^{\frac{\pi}{4}} xsin2xdx$$ (4)

5.2 Given: $y = 2x$ and $y = -x^2 + 6x$

5.2.1 Calculate the coordinates of the points of intersection. (3)

5.2.2 Make a neat sketch to show the enclosed area, the representative strip and the point of intersection. (2)

5.2.3 Calculate the magnitude of the area in QUESTION 5.2.2. (3)

5.2.4 Calculate the volume of the solid of revolution formed when the area in QUESTION 5.2.2 is rotated about the x-axis. (5)

5.3 Calculate the moment of inertia of a circular plate with a radius of 120 cm and thickness of 25 cm about an axis through its centre and perpendicular to the wheel. The mass of the wheel is 30 kg. (3)

[20]

QUESTION 6

6.1 Determine the general solution of:

$$\frac{dy}{dx} = xe^{2x^2}(1 + y)$$ (3)

6.2 Determine the particular solution of:

$$\frac{d^2y}{dx^2} = 8x + \cos 2x, \quad \text{with} \quad \frac{dy}{dx} = 3 \text{ and } y = 2 \text{ at } x = 0$$ (5)

[8]

TOTAL: 100

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MATHEMATICS N5

FORMULA SHEET

Any applicable formula may also be used.

TRIGONOMETRY

$\sin^2 x + \cos^2 x = 1$

$1 + \tan^2 x = \sec^2 x$

$1 + \cot^2 x = \operatorname{cosec}^2 x$

$\sin 2A = 2 \sin A \cos A$

$\cos 2A = \cos^2 A - \sin^2 A$

$$\tan 2A = \frac{2 \tan A}{1 - \tan^2 A}$$

$\sin^2 A = {}^1/_2 - {}^1/_2 \cos 2A$

$\cos^2 A = {}^1/_2 + {}^1/_2 \cos 2A$

$\sin(A \pm B) = \sin A \cos B \pm \sin B \cos A$

$\cos(A \pm B) = \cos A \cos B \pm \sin A \sin B$

$$\tan(A \pm B) = \frac{\tan A \pm \tan B}{1 \pm \tan A \tan B}$$

$\sin A \cos B = {}^1/_2 [\sin(A + B) + \sin(A - B)]$

$\cos A \sin B = {}^1/_2 [\sin(A + B) - \sin(A - B)]$

$\cos A \cos B = {}^1/_2 [\cos(A + B) + \cos(A - B)]$

$\sin A \sin B = {}^1/_2 [\cos(A - B) + \cos(A + B)]$

$$\tan x = \frac{\sin x}{\cos x}; \sin x = \frac{1}{cosec\ x}; \cos x = \frac{1}{\sec x}$$

BINOMIAL THEOREM

$$(x + h)^n = x^n + \mathrm{n}x^{n-1} h + \frac{n(n-1)}{2!} x^{n-2} h^2 + \cdots$$

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DIFFERENTIATION

$$e = -\frac{f(a)}{f'(a)}$$

$$r = a + e$$

PRODUCT RULE

$$y = u(x) \,.\, v(x)$$

$$\frac{dy}{dx} = v \,.\, \frac{du}{dx} + u \,.\, \frac{dv}{dx}$$

$$= v \,.\, u' + u \,.\, v'$$

QUOTIENT RULE

$$y = \frac{u(x)}{v(x)}$$

$$\frac{dy}{dx} = \frac{v \,.\, \frac{du}{dx} - u \,.\, \frac{dv}{dx}}{v^2}$$

$$= \frac{v \,.\, u' - u \,.\, v'}{v^2}$$

CHAIN RULE

$$y = f\big(u(x)\big)$$

$$\frac{dy}{dx} = \frac{dy}{du} \,.\, \frac{du}{dx}$$

$f(x)$	$\frac{d}{dx}f(x)$	$\int f(x)dx$
ax^n	nax^{n-1}	$\frac{ax^{n+1}}{n+1} + c$
a	0	$ax + c$
e^x	e^x	$e^x + c$
a^x	$a^x . lna$	$\frac{a^x}{lna} + c$
$\log_e x$	$\frac{1}{x}$	——

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$f(x)$	$\frac{d}{dx}f(x)$	$\int f(x)dx$
$\log_a x$	$\frac{1}{x\ln a}$	——
$\sin x$	$\cos x$	$-\cos x + c$
$\cos x$	$-\sin x$	$\sin x + c$
$\tan x$	$\sec^2 x$	$ln(\sec x) + c$
$\cot x$	$-\operatorname{cosec}^2 x$	$ln(\sin x) + c$
$\sec x$	$\sec x \,.\tan x$	$ln[\sec x + \tan x] + c$
$\operatorname{cosec} x$	$-\operatorname{cosec} x \,.\cot x$	$ln(\operatorname{cosec} x - \cot x) + c$
$\sin^{-1} x$	$\frac{1}{\sqrt{1-x^2}}$	——
$\cos^{-1} x$	$\frac{-1}{\sqrt{1-x^2}}$	——
$\tan^{-1} x$	$\frac{1}{1+x^2}$	——
$\cot^{-1} x$	$\frac{-1}{1+x^2}$	——
$\sec^{-1} x$	$\frac{1}{\sqrt{x^2-1}}$	——
$\operatorname{cosec}^{-1} x$	$\frac{-1}{x\sqrt{x^2-1}}$	——
$\frac{1}{\sqrt{a^2-x^2}}$	——	$\sin^{-1}\left(\frac{x}{a}\right) + c$
$\frac{1}{a^2+x^2}$	——	$\frac{1}{a}\tan^{-1}\left(\frac{x}{a}\right)+ \mathrm{c}$
$\frac{1}{x\sqrt{x^2+a^2}}$	——	$\frac{1}{a}\sec^{-1}\left(\frac{x}{a}\right) + c$
$\sqrt{a^2-x^2}$	——	$\frac{a^2}{2}\sin^{-1}\left(\frac{x}{a}\right) + \frac{x}{2}\sqrt{a^2-x^2} + c$
$\frac{1}{x^2-a^2}$	——	$\frac{1}{2a}ln\left(\frac{x-a}{x+a}\right) + c$
$\frac{1}{a^2-x^2}$	——	$\frac{1}{2a}ln\left(\frac{a+x}{a-x}\right) + c$

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INTEGRATION

$$\int f(x)\, g'(x)\, dx = f(x)\, g(x) - \int f'(x) g\,(x) dx$$

$$\int \frac{f'(x)}{f(x)} dx = ln\, f(x)\, + c$$

$$\int [f(x)]^n\, f'(x) dx = \frac{[f(x)]^{n+1}}{n\, +1} + c$$

$$\frac{f(x)}{(ax+b)(cx\, +d)} = \frac{A}{ax\, +b} + \frac{B}{cx\, +d}$$

$$\frac{f(x)}{(x\, +a\,)^n} = \frac{A}{(x\, +a)} + \frac{B}{(x\, +a)^2} + \frac{C}{(x+a)^3} + \cdots \frac{Z}{(x\, +a)^n}$$

APPLICATIONS OF INTEGRATION

AREAS

$$A_x = \int_a^b y dx\,; \quad A_x = \int_a^b (y_1 - y_1) dx$$

$$A_y = \int_a^b x dy\,; \quad A_{y=} \int_a^b (x_1 - x_2) dy$$

VOLUMES

$$V_x = \pi \int_a^b y^2\, dx\,; \quad V_x = \pi \int_a^b (y_1^2 - y_2^2) dx$$

$$V_y = \pi \int_a^b x^2\, dy\,; \quad V_y = \pi \int_a^b (x_1^2 - x_2^2) dy$$

SECOND MOMENT OF AREA

$$I_x = \int_a^b r^2\, dA\,; \quad I_{y=} \int_a^b r^2\, dA$$

MOMENT OF INERTIA

$$Mass\, = density\, \times volume$$

$$M\, = \rho v$$

$$DEFINITION: I\, = m\, r^2$$

$$CIRCULAR\ DISC:\ I\, = \frac{mR^2}{2}$$

$$GENERAL: I\, = \int_a^b r^2 dm\, = \rho \int_a^b r^2 dV$$

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